

MPEX Demo · Speaker Notes

9 slides · about 7 minutes · [brackets] = point at or click. Before you start: `python3 serve.py`, open `localhost:8000/` and `/evidence.html`, click **Run All Tests** (17/17, Hub: Ceres).

1. MultiPlanetary Exchange 0:00–0:20

Light takes about four hours to get from here to Neptune. So what does a stock exchange look like when news takes hours to arrive? That's what we built: a physical model of the solar-system network, a market designed around its delays, and an evidence framework that tests the design across scenarios and invariants. Everything you'll see runs on our code. Let's start with the network itself.

2. Delays are computed, not assumed 0:20–1:20

On screen: LIVE: Map · From Earth, To Neptune · Send packet · +1 yr

LIVE. This is our network model running on the real orbits. Pick Earth to Neptune. The planner doesn't use a fixed delay: for every candidate route it sends each hop from where the sender is now to where the receiver will be when the signal lands. It throws out any link that passes too close to the Sun during the flight, and any route through scheduled maintenance; you can see the Relay B routes marked maintenance here. What's left is the best route, through Relay A, and the panel tells you what it costs [point: 4 h 10 m, and 54.76% make it first try]. Send a packet and watch it travel. Now jump ahead one year [click +1 yr]. Different geometry, different routes, different numbers. The market has to live with all of that.

3. Latency ties. Reliability decides. 1:20–2:05

So where should the clearing house sit? Our first instinct was Earth, where most of the money is. The honest answer is that speed doesn't decide it. We reran all 17 scenarios with the hub on Earth and on Ceres, and a cross-planet trade takes essentially the same time either way [point: 38.6 vs 38.5 hours], because Neptune sets the pace. What decides it is reliability over the long run. Over 200 years of orbits, an Earth hub spends far more time hanging on a single relay [point: 8.6% vs 0.8%] and has to re-route far more often [316 vs 4]. Ceres isn't free: it costs about 31% more backbone packets with our Earth-heavy accounts, and its rare blockages can be long. We took that trade.

4. Reserve first, then cross the network 2:05–3:35

On screen: LIVE: Tests & Evidence · Cross-planet trade · replay · Balances before → after

Before the live trade, here's the key idea that makes this market safe [point at the four boxes]. Every planet has its own exchange and ledger, and local trades just clear there. For a cross-planet order, the home exchange locks the cash or shares before anything is sent. So no matter how long the message is in flight, those assets can't be promised to anyone else. Hours of latency can't create a spendable duplicate. The order then joins the batch on Ceres, which waits long enough for the slowest planet. The clearing house on Ceres clears the batch by fixed rules, and the two legs settle back on the home ledgers. LIVE: open Cross-planet trade. Alice on Earth is selling, Eve on Neptune is buying, and you can see both reserved right at the start. Press play on the replay: the packets move across the network and the state steps through to FINAL at the same time. Network and ledger are one system. The batch clears at one fair price [point: \$47.50], Eve's shares are usable on Neptune the moment they land, and Alice's cash follows. Scroll to balances: everything that left one account arrived in the other.

5. Being closer to Ceres doesn't buy priority 3:35–4:15

Being physically closer to Ceres does not buy queue priority. Here's the test. Bob on Earth and Eve on Neptune bid the same price for the last 10 shares. Bob's order gets to Ceres first, about three and a half hours first, because Earth is closer. But Eve's home exchange stamped her order a minute earlier. So Eve gets the shares and Bob's money is released. The rule is simple: best price wins; at the same price, the earlier home-exchange stamp wins; exact ties split pro-rata, with any spare share assigned by a public hash; and everyone pays one midpoint price. When we switched the engine to arrival-time priority by hand, this test failed, so it really does catch the difference.

6. Exactly one financial effect 4:15–4:55

Now we attack the system. In this test we deliberately drop the packet carrying Eve's shares [point at the red LOST box]. The relay waits and tries again, with the same transaction ID, so it's the same transfer, not a new one. Then we send duplicate copies on purpose. They hit the existing record and are ignored. Lost once, retried once, duplicated twice, and exactly one financial effect: Eve gets her shares once. And after every single step the engine checks that no money or shares were created, lost or spent twice.

7. The winner is paid in full 4:55–5:40

On screen: LIVE: Funded default · Futures · MOI-F300 · lower chart · Settlement moments

Second attack: a futures trader who stops paying. LIVE: open Funded default. Eve is long futures from Neptune. Because she's far away, she has to post more margin up front; the clearing house needs to survive a longer silence from her. Then the price falls hard [point at the lower chart, where the requirement line crosses her collateral]. A margin call goes out, and she never answers. The clearing house closes her position at the next official price. Her margin runs out, and the guarantee fund covers the rest. The winner is paid in full [point: \$22,000], with zero unbacked, in this tested scenario. And that fund was contributed by members after hour zero, not created from nothing.

8. Not cherry-picked: 200 years of orbits 5:40–6:30

Fair question at this point: did we just pick examples that work? So we tested the design across changing orbits. Each tick on this chart is a time when the Sun blocks one settlement's link to a relay, over 200 years. Our scan found 5,748 of those blockage starts and ends, and refined every one to the second; at each one the link flips exactly where predicted. In the 200-year scan, no settlement loses both relays simultaneously. We also moved the start date forward 1, 10 and 100 years and the batch still runs 31 to 34 hours under the same rules. We cut Ceres itself off for 72 hours: local trading carries on, and service recovers. And underneath, the dashboard: all 17 test scenarios pass. Our evidence set adds one more futures run with a rising price, and all 18 evidence runs conserve cash and shares at every step.

9. A fair, solvent market across the solar system 6:30–7:00

So, what did we demonstrate? All 17 test scenarios pass on real orbits, and all 18 evidence runs, the tests plus one rising-price futures run, conserve cash and shares at every step. No double spends, no duplicate settlements. In our default test, the winner is still paid in full. And putting the clearing house on Ceres cuts single-relay exposure roughly tenfold at the same speed. One thing we haven't built yet, and we say so in the paper: moving sessions to the other relay ahead of a long blockage is designed, not implemented, so at one difficult epoch a funded future stays open past maturity. It stays funded; nothing is lost. Across the implemented system, the physics, routing, market, settlement, default handling and evidence all run together on the same model. Thank you.